

Nested Multi-Timescale Learning and Control for Voltage-Aware EV Charging Coordination

Abraheem Rashid, Faisal Iradat, Waseem Iqbal, and Yawar Abbas Bangash

Abstract—Residential electric-vehicle (EV) charging concentrates in the evening, when energy is expensive and feeder voltages are lowest. Prices change hourly, dispatch runs every few minutes and voltages must be corrected within a minute, yet most learned coordinators act at one rate. We propose a nested controller with one level per timescale. Proximal policy optimization sets an hourly price corridor. A shared deep deterministic policy gradient agent dispatches every 15 min as a learned residual on a deadline-aware cheapest-slot plan, and least-laxity-first allocation keeps dispatch within charger and transformer limits. A non-learned layer corrects voltage every 60 s with charger reactive power bounded by the inverter rating. On IEEE 33- and 69-bus feeders with AC power flow, real prices and household loads, a synthetic residential fleet and, for generalization, recorded workplace sessions (50 held-out days), the controller costs 30.8% less than uncoordinated charging at 80% penetration without voltage violations and is the cheapest learned controller, but 4.1% more than a price-aware heuristic. Ablations attribute the cost gains to the planning prior and the absence of violations to the voltage layer. Attached to any baseline, it removes all violations up to 100% penetration whenever the base load alone respects the floor.

Index Terms—Deep reinforcement learning, distribution networks, electric vehicle charging, multi-timescale control, reactive power, residual policy learning, transportation electrification, voltage regulation.

I. INTRODUCTION

TRANSPORTATION electrification is accelerating: global electric-car sales exceeded 17 million in 2024 and are expected to pass 20 million in 2025 [1]. Residential charging of electric vehicles (EVs) concentrates in the evening, when drivers return home and household demand is already at its peak. On weak radial feeders this added load depresses end-of-line voltages toward and below the 0.95 p.u. lower limit of the ANSI C84.1 service range [2]–[4], and it buys energy in the most expensive hours of the day-ahead market. Coordinated charging can avoid both, because most vehicles stay plugged in far longer than they need to charge.

Existing coordinators fall into three families. *Optimization-based* schedulers [3], [5] reach good schedules when the model and the forecasts are accurate, but they need vehicle-level information over the whole horizon, network-aware variants

also need a network model, and they are usually solved on a single decision interval. *Decentralized* schemes such as time-of-use (TOU) tariffs, game-theoretic demand response and priority rules [6]–[8] scale well but do not observe the network. Deep *reinforcement learning* (RL) [9]–[17] learns from data, but most agents are flat: one policy acts at one rate, and voltage limits, where they are modeled, enter as reward penalties or expected-cost constraints that are not enforced at run time.

The underlying problem is a timescale mismatch. Wholesale prices change every hour, aggregators re-dispatch every few minutes, and voltages respond within seconds. *Nested learning* [18] views a learning system as a set of interconnected optimization problems, each updated at its own frequency. We use it as a design principle: one level per timescale, with set points passed down and measurements passed up.

This paper makes four contributions.

- 1) **A nested controller** (Section IV) that matches each level to its timescale: an hourly price-corridor policy trained with proximal policy optimization (PPO), a 15-min dispatch policy trained with deep deterministic policy gradient (DDPG) and shared by all aggregators, and a non-learned voltage layer that acts every 60 s.
- 2) **Plan-residual dispatch with feasibility by construction**. Level 2 learns a correction to a deadline-aware cheapest-slot plan (residual policy learning [19], [20]), so training starts from a sound schedule instead of from scratch. Least-laxity-first (LLF) allocation [8] with a deadline guard maps the aggregate set point to per-vehicle rates that satisfy charger and transformer limits exactly.
- 3) **Reactive voltage correction bounded by the inverter rating**. Level 3 bounds each charger’s reactive power by $Q_i^{\max} = \sqrt{S_i^2 - P_i^2}$ [21], so curtailing active power never removes reactive headroom. A secant step on the measured voltage sensitivity closes the voltage gap, and curtailment is used only when the headroom is exhausted. The layer is independent of the policy above it, and we evaluate it with every baseline as well.
- 4) **A reproducible, paired evaluation** (Sections III and V) with AC power flow on two IEEE feeders; real ENTSO-E prices [22], Pecan Street household loads [23] and ACN-Data sessions [24]; 50 held-out days; five training seeds; and eight baselines, including model-predictive control (MPC) with a linear-programming optimal power flow (LP-OPF) and perfect foresight, PPO-Lagrangian, constrained policy optimization (CPO) and a goal-conditioned hierarchical RL (HRL) agent. One command regenerates every number, table and figure in

(Corresponding author: Waseem Iqbal.)

A. Rashid is with the Department of Computer Science, Brunel University London, London, U.K. (e-mail: 2546393@brunel.ac.uk).

F. Iradat is with the Department of Computer Science, Institute of Business Administration, Karachi, Pakistan (e-mail: firadat@iba.edu.pk).

W. Iqbal is with the Department of Electrical and Computer Engineering, Sultan Qaboos University, Muscat, Oman (e-mail: m.waseem@squ.edu.om).

Y. A. Bangash is with the School of Computing and Artificial Intelligence, Sunway University, Selangor, Malaysia (e-mail: yawarb@sunway.edu.my).

this paper from the public code.

II. RELATED WORK

Optimization. Clement-Nyns *et al.* [3] schedule plug-in vehicles with quadratic and dynamic programming to limit losses and voltage deviations on a residential feeder, and Ortega-Vazquez [5] schedules charging and vehicle-to-grid services at household level, including battery degradation and price uncertainty. Such schedulers need forecasts of arrivals, departures and energy needs; we include a model-predictive LP-OPF with *perfect* foresight as a strong reference.

Decentralized coordination. Mohsenian-Rad *et al.* [6] formulate demand-side management as a game, and Ma *et al.* [7] show that decentralized charging control can fill the overnight valley. Xu *et al.* [8] show that schedules can be improved by giving priority to vehicles with less laxity and longer remaining charging time; our least-laxity-first allocation follows this principle.

Learning-based charging. Deep RL has scheduled real-time charging with deep Q-learning and price features [9] and with DDPG [10], exploited the structure of optimal policies in a distribution network [11], coordinated multiple agents and navigation [12]–[14], added vehicle-to-grid operation and degradation [15], controlled a fleet [16], scaled with graph neural networks [17] and priced charging stations [25]. EV2Gym [26] and SustainGym [27] provide simulation benchmarks and, for this study, the bundled open data. Saner *et al.* [28] propose a cooperative hierarchical multi-agent system for several stations, and Wang *et al.* [29] regulate voltage on two timescales for flash-charged electric buses. Hierarchical RL in general is surveyed in [30].

Safe RL and voltage control. CPO [31] and Lagrangian methods treat safety as an expected-cost constraint; their use in power systems is reviewed in [32], [33]. Safety-constrained deep RL has been applied to Volt-VAR control [34], [35], active distribution networks [36], real-time voltage control with stability guarantees [37] and multi-agent active voltage control [38]. Multi-agent deep RL also coordinates Volt-VAR devices [39] and active and reactive power [40], and physics-informed graph representations make voltage control robust to bad measurements [41]. EV chargers have supported feeder voltage with vehicle-to-grid active power [42], [43], with reactive power [21], [44] and with coordinated active and reactive control [45]; inverter reactive capability is standardized for distributed energy resources in IEEE Std 1547-2018 [46].

Positioning. Table I summarizes the gap. Learned charging coordinators act at one decision rate and either do not model voltage or learn it through rewards or constraints; optimization needs a network model and forecasts of all vehicles. Our controller separates the timescales, keeps feasibility out of the learning problem and corrects voltage on the measured state, using only day-ahead prices, the states of the plugged-in vehicles and measured bus voltages.

III. SYSTEM MODEL

A. Feeder and Power Flow

We use the IEEE 33-bus [47] and 69-bus [48] radial feeders with the MATPOWER data [49] (open tie lines removed). Five

TABLE I
POSITIONING BY METHOD CLASS (RL: REINFORCEMENT LEARNING; –: NOT CHARACTERIZED HERE)

Class	Time-scales	Voltage handling	Charging limits
Network optimization [3]	one	network model	constraints
Household scheduling [5]	one	not modeled	constraints
Decentralized [7], [8]	one	not modeled	per vehicle
Deep RL charging [9], [10]	one	not modeled	action bounds
Safe RL [31], [34]	one	expected cost	–
Two-timescale deep RL [29]	two	deep RL	–
Nested (ours)	three	measured, corrected	by construction

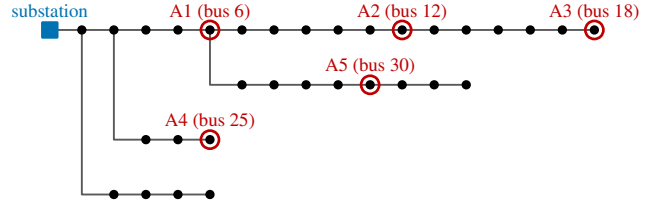


Fig. 1. IEEE 33-bus feeder (MATPOWER data, open ties removed) with the five aggregator connection points A1–A5.

aggregators connect at buses 6, 12, 18, 25 and 30 of the 33-bus feeder (9, 21, 35, 49 and 61 on the 69-bus feeder), at electrically strong and weak locations (Fig. 1); vehicles are assigned to them at random in proportions 0.18, 0.18, 0.19, 0.19 and 0.26. Aggregator k serves its vehicles through a transformer of rating $P_k^{\text{cap}} = 500 \text{ kW}$. With constant-power loads the radial AC power-flow equations are

$$\mathbf{V} = V_0 \mathbf{1} - \mathbf{Z}^{\text{path}} (\mathbf{S} \oslash \mathbf{V})^*, \quad (1)$$

where $\mathbf{V}$ holds the complex voltages and $\mathbf{S}$ the complex loads of the non-substation buses, $V_0 = 1.03 \text{ p.u.}$ is the substation voltage, $\mathbf{1}$ is the all-ones vector, $\oslash$ is element-wise division, $(\cdot)^*$ is element-wise complex conjugation, and Z_{nm}^{path} is the impedance of the path shared by buses n and m from the substation. We solve (1) by the backward/forward sweep [50] to 10^{-10} p.u. ; our tests match pandapower’s Newton–Raphson solution [51] to 10^{-8} p.u. , and a power flow is solved every 60 s of simulated time.

B. Data

Prices. Hourly Netherlands day-ahead prices from the ENTSO-E Transparency Platform [22], as distributed with EV2Gym [26], converted to local standard time: 2023 for training, 2024 for testing, and 2019 as an alternative price regime.

Household load. One year of 15-min consumption of 25 Pecan Street homes [23], as distributed with EV2Gym [26], averaged and normalized by its 99.5th percentile. The profile scales the nominal feeder loads, matched by day of year (so training and test days share it), with multiplicative noise of standard deviation 0.02 at every power-flow step. The base-load scale is calibrated as the largest value (in steps of 0.01) for which the feeder *without* EVs stays at or above 0.955 p.u.

over the whole year (0.78 for the 33-bus and 0.75 for the 69-bus feeder). The voltage problems studied here are therefore caused by EV charging.

Vehicles. The main experiments use a synthetic residential fleet whose parameters are stated assumptions: arrivals $\mathcal{N}(18, 2^2)$ h clipped to 12:00–23:30, dwell $\mathcal{N}(12, 3^2)$ h clipped to 2–16 h, initial and target state of charge $\mathcal{N}(0.35, 0.15^2)$ and $\mathcal{N}(0.85, 0.10^2)$ (clipped to $[0.05, 0.9]$ and to at least 0.05 above the initial value), batteries of 40, 60, 75 or 100 kWh (equally likely), and chargers of rating $\bar{c} = 11$ kW with efficiency $\eta = 0.95$. Episodes run 24 h from 12:00. For generalization we also use real sessions from ACN-Data [24] at Caltech and JPL (bundled with SustainGym [27]): each sampled vehicle keeps a recorded session's arrival time, dwell and delivered energy (2019 for training, 2020–2021 for testing, weekdays, episodes from 06:00). Departures are truncated at the end of the episode and requests capped at what the charger can deliver during the dwell, so full service is attainable except for the wait (under 15 min) for the first dispatch after arrival. EV penetration is the number of vehicles per household, with 300 households behind the five aggregators.

C. Driver Response

Price response is part of the environment, not a control level. Each vehicle accepts the execution price p_k of its aggregator with the Fishbein–Ajzen attitude model [52],

$$P_i^{\text{acc}} = \sigma \left(w_i^c \frac{\lambda^{\text{ref}} - p_k}{\lambda^{\text{ref}}} + w_i^n \bar{A} + b_i \right), \quad (2)$$

with logistic σ , reference rate $\lambda^{\text{ref}} = \text{€}0.20/\text{kWh}$, last-interval fleet acceptance $\bar{A}$ (floored at 0.3) and heterogeneous weights $w_i^c \sim \mathcal{N}(5, 1)$, $w_i^n \sim \mathcal{N}(2, 0.5^2)$, $b_i \sim \mathcal{N}(0, 0.5^2)$. A driver whose remaining need exceeds 0.7 times the charger rating times the hours to departure accepts regardless of price. Acceptance is drawn once per dispatch interval.

D. Energy Accounting and Metrics

A vehicle draws power only while it is plugged in, has accepted the current price and still needs energy, and it never draws more than completes its need within a step. The energy cost is the day-ahead price times the energy actually drawn (after any curtailment). We report the daily cost (€), the cost per delivered kWh, the service quality (SQ, delivered over requested energy), the violation rate (Viol., share of 60-s steps with any bus below $\underline{V} = 0.95$ p.u. after any correction), the mean daily minimum voltage (the lowest bus voltage $V^{\min}$ of each day, averaged over days; in general, $V^{\min}$ is the lowest bus voltage of the period considered), the curtailed energy (Curt.) and the unmet energy (need left when vehicles depart), both in kWh per day, and the mean retail price paid by drivers (Retail).

IV. NESTED CONTROLLER

Fig. 2 shows the architecture. Each level is a decision process at its own rate whose environment includes the drivers' price response and the closed-loop behavior of the faster levels.

A. Level 1: Price Corridor (PPO, Hourly)

The state $s_t^{(1)} \in \mathbb{R}^{11}$ holds the mean bus voltage, the substation active and reactive power, the current day-ahead price, the share of vehicles plugged in, their mean state of charge, the line-loading margin (assumed rating 0.4 kA per line), the substation loading (assumed 8 MVA), the hour (sine and cosine) and the mean price over the next 12 h. The action $a_t^{(1)} \in [0, 1]^2$, drawn from Beta distributions, sets the center c_t and width w_t of the corridor $[p_t^{\min}, p_t^{\max}] = [c_t - w_t/2, c_t + w_t/2]$ (clipped to $[p^{\text{lo}}, p^{\text{hi}}]$):

$$c_t = \lambda^{\text{ref}} + (2a_{t,1}^{(1)} - 1) \begin{cases} \lambda^{\text{ref}} - p^{\text{lo}}, & a_{t,1}^{(1)} < 0.5, \\ p^{\text{hi}} - \lambda^{\text{ref}}, & \text{otherwise,} \end{cases} \\ w_t = w^{\min} + a_{t,2}^{(1)} (w^{\max} - w^{\min}), \quad (3)$$

with bounds $p^{\text{lo}} = \text{€}0.05$ and $p^{\text{hi}} = \text{€}0.40/\text{kWh}$ and widths $w^{\min} = \text{€}0.02$ and $w^{\max} = \text{€}0.20/\text{kWh}$. The neutral action $(0.5, 0.5)$ gives the *neutral corridor* $[\text{€}0.145, \text{€}0.255]/\text{kWh}$, centered on the reference rate, so an untrained policy, whose mean action is close to it, charges about the flat tariff of the baselines. The reward of hour t ,

$$r_t^{(1)} = -\frac{C_t}{100} - 50(20\Delta V_t)^2 - 10\mathbb{I}[\Delta V_t > 0] \\ - 5\frac{U_t}{100} - 0.5\frac{X_t}{100} + \omega\frac{R_t - \lambda^{\text{ref}}E_t^{\text{del}}}{100}, \quad (4)$$

penalizes energy cost C_t , the voltage deficit $\Delta V_t = \max(0, \underline{V} - V_t^{\min})$ with $V_t^{\min}$ the lowest bus voltage of the hour, unmet energy U_t of departing vehicles and curtailment X_t ($\mathbb{I}$: indicator); the last term, with retail revenue R_t and delivered energy E_t^{del} , is the revenue-neutrality term defined below. PPO [53] trains the policy with clipping 0.2, $\gamma = 0.99$ and generalized advantage estimation (GAE) with $\lambda_{\text{GAE}} = 0.95$.

B. Level 2: Plan-Residual Dispatch (DDPG, 15 min)

State. The state of aggregator k , $s_{\tau,k}^{(2)} \in \mathbb{R}^{28}$, holds its bus voltage, the hour, the current price, the day-ahead prices of the current and the next 11 hours, the corridor, the number of plugged-in vehicles, their remaining energy need (total, and split into four laxity bins: below 1, 1–3, 3–6 and above 6 h), the power they can absorb, their mean remaining dwell, the last fleet acceptance rate, and the plan set point $u_{\tau,k}^0$ defined below. One actor μ and one critic are shared by the five aggregators, with a one-hot aggregator identifier appended to the state (parameter sharing [54]).

Planning prior. For each plugged-in vehicle i with remaining need e_i , the plan selects the $\lceil e_i/(\eta\bar{c}\Delta) \rceil$ cheapest intervals before departure ($\Delta = 15$ min); prices beyond the 12-h look-ahead are valued at the mean known price. The plan set point $u_{\tau,k}^0 \in [0, 1]$ is the power of the vehicles whose current interval is selected, as a fraction of the power the aggregator can deliver now. The plan uses only information the aggregator has: the price look-ahead and the needs and departure times of its plugged-in vehicles.

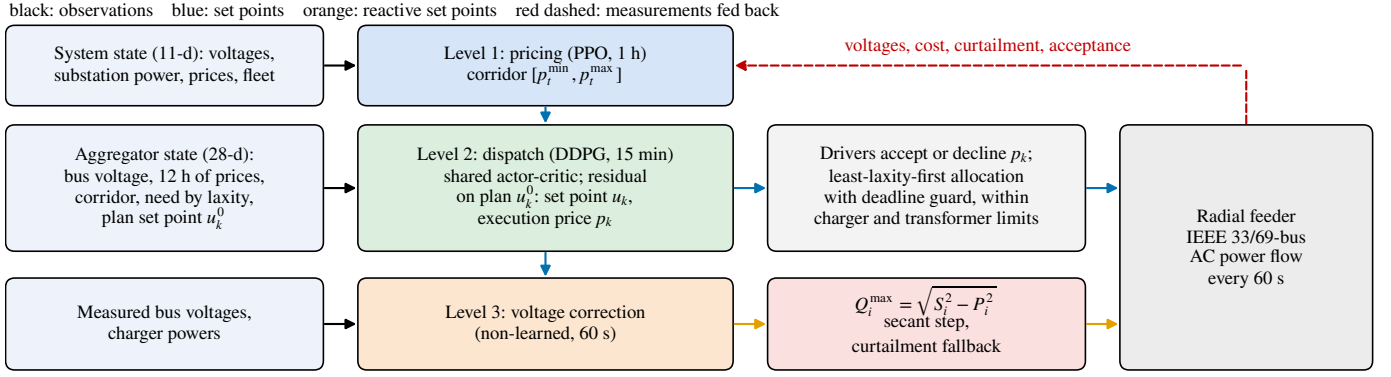


Fig. 2. Nested controller. Black arrows: observations; blue: set points (top-down); orange: reactive set points computed by Level 3 from measured voltages; red, dashed: measurements fed back (bottom-up).

Action. The actor outputs $a_{\tau,k} = \mu(s_{\tau,k}^{(2)}) \in [0, 1]^2$. The first component corrects the plan, and the second sets the execution price inside the corridor:

$$u_{\tau,k} = \text{clip}(u_{\tau,k}^0 + \rho(2a_{\tau,k,1} - 1), 0, 1), \quad (5)$$

$$p_{\tau,k} = p_t^{\min} + a_{\tau,k,2}(p_t^{\max} - p_t^{\min}), \quad (6)$$

so an actor output of $a_{\tau,k,1} = 0.5$ reproduces the plan; the residual scale ρ is listed in Table II.

Feasibility by construction. Let $\hat{c}_i = \min\{\bar{c}, e_i/(\eta\Delta)\}$ for the vehicles that accepted the price. The set point dispatches $P_{\tau,k} = u_{\tau,k} \min\{P_k^{\text{cap}}, \sum_i \hat{c}_i\}$, which is filled in order of increasing laxity $\ell_i = h_i - e_i/(\eta\bar{c})$ (h_i : hours to departure) [8]. A deadline guard raises $P_{\tau,k}$ to cover every vehicle with $\ell_i \leq \Delta$, up to the same limit. Every dispatch therefore lies in

$$\mathcal{C}_k = \left\{c : 0 \leq c_i \leq \hat{c}_i, \sum_i c_i \leq P_k^{\text{cap}}\right\}. \quad (7)$$

Methods that output per-vehicle rates are mapped onto $\mathcal{C}_k$ by exact Euclidean projection (bisection on the dual of the sum constraint).

Reward. The reward of aggregator k in interval τ is

$$r_{\tau,k}^{(2)} = -C_{\tau,k} + \omega(p_{\tau,k} - \lambda^{\text{ref}})E_{\tau,k}^{\text{del}} - 2U_{\tau,k} - 0.5G_{\tau,k} - 0.5X_{\tau,k}, \quad (8)$$

where $C_{\tau,k}$ is the energy cost of the interval (summed over a day it is exactly the daily cost), $U_{\tau,k}$ the unmet energy of departing vehicles, $G_{\tau,k}$ the energy that can no longer be delivered before departure and $X_{\tau,k}$ the curtailed energy (energies in kWh, money in €). Because one day spans 96 decisions, we use $\gamma = 0.999$; with $\gamma = 0.99$ a cost incurred 12 h later would be discounted by almost 40% ($0.99^{48} \approx 0.62$).

Revenue-neutral pricing. A profit-seeking aggregator would raise retail prices, which lowers acceptance and pushes charging into expensive hours. We instead require the mean retail price paid to equal the flat reference rate λ^{ref} charged by the non-pricing baselines. The weight ω in (4) and (8) is the Lagrange multiplier of this constraint. Plain dual ascent

oscillated in our experiments, so after every training episode we use a proportional–integral (PI) Lagrangian update [55],

$$\varepsilon = (\lambda^{\text{ref}} - \bar{p})/\lambda^{\text{ref}}, \quad \bar{\varepsilon} \leftarrow (1 - \alpha)\bar{\varepsilon} + \alpha\varepsilon, \\ \omega = \text{clip}\left(K_p \bar{\varepsilon} + K_i \sum \varepsilon, \omega^{\min}, \omega^{\max}\right), \quad (9)$$

where $\bar{p}$ is the mean retail price paid in the episode, $\bar{\varepsilon}$ is an exponential moving average (EMA) with weight α , and the sum runs over past episodes and is clipped against wind-up (gains and bounds in Table II). The multiplier enforces the constraint on average over the training episodes; the final deterministic policy can still charge slightly more or less than λ^{ref} . After training we therefore calibrate the tariff level: a constant offset added to every execution price (the sum clipped to $[p^{\text{lo}}, p^{\text{hi}}]$, so it may leave the corridor by up to the offset) is found by bisection such that, on ten randomly drawn 2023 training days, the deterministic policy's mean retail price paid equals λ^{ref} . The same calibration is applied to every learned method, uses only training data and moved prices by at most €0.033/kWh. Prices can thus shift charging in time while drivers pay about the reference rate.

DDPG [56] trains the actor and critic (256–128–64 units) on transitions $(s_\tau, a_\tau, r_\tau, s_{\tau+1})$, where a_τ is the action *actually applied* after the guard, mapped back through (5) and clipped to $[0, 1]$. The actor's output layer is initialized near zero, so training starts from the plan.

C. Level 3: Reactive Voltage Correction (60 s)

Level 3 runs inside every 60-s step (Fig. 3). After the power flow, if $\min_n |V_n| < \underline{V} + \delta$ (with $\delta = 0.001$ p.u.), the plugged-in chargers at aggregator bus k offer the headroom

$$\bar{Q}_k = \sum_{i \in k} \sqrt{\max\{S_i^2 - P_i^2, 0\}}, \quad S_i = 12 \text{ kVA}. \quad (10)$$

At the full charging power of 11 kW a charger thus still offers 4.8 kvar, at $P_i = 0$ it offers its full rating, and curtailing active power *increases* reactive headroom. In our model S_i bounds only the reactive capability; the 11-kW active-power limit is separate. The first injection is proportional to the voltage gap, saturates when the lowest voltage $V^{\min}$ reaches 0.90 p.u., and is weighted toward the most depressed aggregator buses.

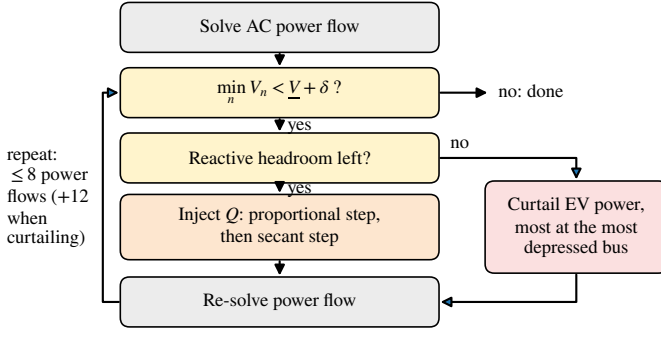


Fig. 3. Level-3 correction inside one 60-s step.

Algorithm 1 One 15-min Interval of the Nested Controller

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1: if hour boundary then observe  $s^{(1)}$ ; set corridor by (3)
2: for each aggregator  $k$  do
3:   observe  $s_k^{(2)}$  (incl. plan  $u_k^0$ );  $a_k \leftarrow \mu(s_k^{(2)})$ 
4:   set point  $u_k$  by (5); price  $p_k$  by (6)
5: end for
6: drivers accept or decline by (2) (environment)
7: LLF allocation with deadline guard onto  $C_k$  (7)
8: for each of the fifteen 60-s steps do
9:   solve the AC power flow (1)
10:  if  $V^{\min} < \underline{V} + \delta$  then Level-3 loop (10), (11)
11:  book drawn energy, cost and curtailment
12: end for
13: (training) store  $(s, a^{\text{applied}}, r, s')$ ; update the DDPG agent
14: (training, end of day) update the PPO policy and  $\omega$  by (9)

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Each later iteration measures the sensitivity $\hat{s} = \Delta V^{\min} / \Delta Q$ between the last two power-flow solutions and, if ΔQ and $\Delta V^{\min}$ are both positive, takes the secant step

$$\Delta Q = 1.2 (\underline{V} + \delta - V^{\min}) / \hat{s}, \quad (11)$$

split across buses by voltage depression and capped by the remaining headroom; otherwise the proportional step is repeated. If no headroom is left, EV power is shed in steps of up to 20% per iteration, most at the most depressed bus, and the flow is re-solved. The loop stops when $V^{\min} \geq \underline{V} + \delta$ or when the iteration limit is reached: eight corrective power flows, raised to twenty once curtailment has started. Because the sensitivity is measured on successive AC solutions, the step needs no precomputed network model. Level 3 does not depend on the policy above it; we therefore also evaluate it with every baseline (“+L3”).

D. Training

Episodes are random 2023 days. A curriculum raises the fleet from 60 to 300 vehicles in steps of 60 and advances when the last 20 episodes reach mean SQ ≥ 0.95 with mean curtailment ≤ 20 kWh; the final stage draws the fleet size uniformly between 120 and 300. Each run has 600 episodes and takes about 0.5 h on a single CPU core. Algorithm 1 summarizes one dispatch interval, and Table II lists the hyperparameters, read directly from the configuration file and the code.

V. EXPERIMENTAL SETUP**A. Scenarios**

S1, S2, S3 and S5 use 120, 200, 240 and 300 vehicles (40%–100% penetration). S4 repeats S3 with an hourly-persistent

TABLE II
HYPERPARAMETERS, READ FROM THE CONFIGURATION FILE AND THE CODE

Level 1 (PPO)	actor/critic 256–128–64, Beta policy, lr 0.0003, γ 0.99, λ_{GAE} 0.95, clip 0.2, 10 epochs per update, entropy 0.001
Level 2 (DDPG)	actor/critic 256–128–64, shared, one-hot id lr actor/critic 0.0001/0.001, γ 0.999, τ 0.005 batch 128, buffer 200 000, warm-up 1 000 transitions Gaussian action noise, std 0.3→0.05 (factor 0.995 per episode); residual scale ρ 0.1
Multiplier	K_p 2.0, K_i 0.05, EMA weight α 0.2, $\omega \in [-2.0, 2.0]$, $ K_i \sum e \leq 2.0$
Training Baselines	600 episodes; curriculum window 20 same networks, learning rates, episodes and curriculum; safe RL: γ 0.999, cost limit 0, dual step 0.05 (PPO-Lag.), KL radius 0.01 (CPO)
Level 3	δ 0.001 p.u., S_i 12.0 kVA, ≤ 8 (+12 when curtailing) power flows

deviation of the base load from its profile (standard deviation 15%, clipped to $\pm 40\%$); it is a forecast error for the LP-OPF, the only method that uses a load forecast. Two stress tests probe Level 3. S6 forbids all charging between 19:00 and 20:00, so that charging resumes in a synchronized rebound. S7 has 360 vehicles (120%) and emulates reduced reactive capability by halving the inverter rating to $S_i = 6$ kVA: a charger drawing more than 6 kW then offers no reactive power.

B. Baselines

Rule-based. Uncoordinated charging (full power on arrival); a TOU timer (off-peak 23:00–07:00, earlier only when laxity is exhausted); and the price-aware heuristic, which charges each vehicle at full power in its planned cheapest intervals, that is, the Level-2 plan executed per vehicle, with the same 12-h price look-ahead. *Model-predictive LP-OPF.* At every interval an LP over the rest of the day (HiGHS [57]) minimizes energy cost plus a penalty of €5/kWh on unmet energy, subject to per-vehicle energy, charger and transformer limits and to linearized voltage limits ($\geq \underline{V} + \delta$ at every bus) from finite-difference sensitivities of the AC power flow at the forecast base-load (no-EV) operating point of each interval. It has *perfect foresight* of prices and of every vehicle’s arrival, departure and need, which the asterisk in LP-OPF* marks in tables and figures. It is a strong model-based reference rather than a deployable controller; it is not a strict bound, because its voltage model is linearized and drivers may still decline. *Learned.* Flat DDPG (one agent, all aggregators’ states and actions), PPO-Lagrangian and CPO [31] (expected-cost constraint on violations, limit $d = 0$), and a goal-conditioned HRL agent (hourly DDPG goals, 15-min DDPG workers). They observe the same states, including the plan set point, use the same allocation layer, reward terms, discount factor and revenue-neutral multiplier, and set execution prices inside the neutral corridor. Because they are trained without Level 3, Flat DDPG and the HRL agent add a voltage penalty to the reward (the hierarchical workers also receive a goal-tracking term), and PPO-Lagrangian and CPO constrain it as the cost. Each is also evaluated with Level 3 attached. Uncoordinated

charging, the price-aware heuristic and the LP-OPF charge the flat reference rate λ^{ref} ; the TOU timer charges its three-tier tariff (€0.12/kWh 23:00–07:00, €0.30/kWh 17:00–21:00 and €0.20/kWh otherwise).

C. Protocol

All methods are evaluated on the same 50 distinct 2024 days, with identical fleets and noise for each day (paired design). Learned methods are trained with five seeds (three in the ablations and generalization tests); each day's metric is averaged over seeds. We report means with 95% bootstrap confidence intervals over days (10 000 resamples) and compare methods with Wilcoxon signed-rank tests on the paired daily differences, Holm-corrected across the comparison family. The allocation rule was selected before testing, on 30 held-out 2023 days with a criterion fixed in advance: over three variants that serve planned slots first (two seeds each), the S3 cost fell by up to 1.3%, but 24%–89% (S3) and 39%–103% (S7) more energy was left undelivered, so least-laxity-first allocation was kept.

VI. RESULTS

A. Main Comparison

Table III compares all methods at S3. Uncoordinated charging costs €931.4 per day and violates the voltage floor in 3.99% of the 60-s steps. Attaching Level 3 removes every violation of the rule-based methods at S1, S2, S3 and S5 and changes their cost by at most 0.10%, because the correction uses reactive power and curtails only when the headroom is exhausted. The perfect-foresight LP-OPF is the cheapest method (€612.0). Among the methods without perfect foresight, the price-aware heuristic is cheapest (€619.8). The nested controller costs €645.0, which is 30.8% less than uncoordinated charging and 4.1% more than the price-aware heuristic, with no violations, a service quality of 0.997 and a mean retail price of €0.201/kWh. It is the cheapest learned controller, 13.0% below the best learned baseline (Hierarchical RL + L3), and the least sensitive to the training seed: the standard deviation of its mean daily cost over seeds is €3.1, against €11–54 for the learned baselines.

B. Penetration and Base-Load Deviation

Table IV and Fig. 4 extend the comparison to S1–S5. Without Level 3, violations of uncoordinated and TOU charging grow with penetration, up to 6.82% and 7.55%. Without base-load deviation (S1, S2, S3, S5) the nested controller and every “+L3” method record no violations. S4 is different: with the base load deviating from its profile, the household load alone breaks the floor on 4 of the 50 days even when no vehicle charges. Level 3 removes most of these violations with the reactive headroom of idle chargers, which offer their full rating at zero charging power; the remainder, at most 0.005% of the steps for the nested controller, occurs only on those days. At every penetration level the nested controller costs 30.6%–31.0% less than uncoordinated charging and 3.7%–4.3% more than the price-aware heuristic, and it stays below every learned baseline by at least 12.6%.

C. Statistical Comparison

Table V tests the daily differences at S3 against the baselines with Level 3 attached, so that both sides of each comparison include the voltage layer. Without it, the mean differences change by at most €0.7 and the same cost differences are significant. The cost differences to uncoordinated charging and to every learned baseline are significant after Holm correction, and so is the cost premium relative to the price-aware heuristic and the LP-OPF. Service-quality differences are at most 0.005 in absolute terms; the nested controller's service quality is significantly lower only than that of uncoordinated charging and CPO.

D. Level 3 Under Stress

Table VI probes Level 3 beyond nominal conditions. In S6 no charging is allowed for an hour in the evening peak, after which charging resumes at once; every controller with Level 3 in Table VI keeps the floor. S7 exceeds what the reduced reactive capability can correct: all methods except the perfect-foresight LP-OPF keep some violations, and the methods separate. The nested controller curtails 67% less energy than the price-aware heuristic, leaves 64% less energy undelivered and has 92% fewer violations; its service quality is 0.997 against 0.991. The price of this robustness is energy cost: €982.9 against €940.4 per day. Table VII shows that the two advantages have different sources.

E. Contribution of the Learned Levels

Table VII separates what each learned level contributes, with Level 3 active throughout. Executed without learning, the plan costs 1.3% more at S3 than the per-vehicle price-aware heuristic, because the aggregate set point is filled least-laxity-first rather than by each vehicle's own cheapest slots. Relative to the plan, learned prices alone raise the S3 cost by 1.0%, learned dispatch alone by 2.8%, and the full controller by 2.7%: at nominal load, learning does not lower the cost. Under stress, the lower unmet energy of the nested controller needs no learning. The plan executed least-laxity-first already leaves 65% less energy undelivered than the per-vehicle heuristic in S7, and the full controller leaves about as much undelivered as the plan (38.2 against 37.0 kWh per day). The lower curtailment comes from learning: the full controller curtails 60% less energy than the plan, learned dispatch alone 38% less and learned prices alone 28% less.

F. Ablations

Table VIII removes one component at a time. With three seeds per configuration, we call a cost difference resolved when the per-seed mean costs do not overlap with those of the complete framework. The planning prior is resolved at S3 and S5: learning the dispatch from scratch raises the cost by 7.9% and 7.3%. Removing the deadline guard is resolved only at S5 (+1.0%). Proportional allocation, a single timescale (Level 1 acting every 15 min) and a fixed neutral corridor instead of Level 1 change the cost by at most 2.8%, within the seed spread; without Level 1 the S3 cost is even 0.5% lower, so we

TABLE III

RESULTS AT S3 (240 VEHICLES, 80% PENETRATION; IEEE 33-BUS; 50 HELD-OUT DAYS; LEARNED METHODS AVERAGED OVER FIVE TRAINING SEEDS). COST: MEAN DAILY COST WITH 95% BOOTSTRAP CONFIDENCE INTERVAL; COST/KWH AND RETAIL IN €/KWH; OTHER METRICS AS DEFINED IN SECTION III-D; +L3: LEVEL 3 ATTACHED

Method	Cost (€)	Cost/kWh	Retail	SQ	Viol. (%)	Mean $V^{\min}$	Curt. (kWh)
Uncoordinated	931.4 [833.8, 1060.7]	0.1146	0.200	1.000	3.99	0.9554	0.0
Uncoordinated + L3	931.4 [835.0, 1062.6]	0.1146	0.200	1.000	0.00	0.9615	5.0
TOU timer	697.7 [639.6, 751.2]	0.0863	0.132	0.995	4.30	0.9372	0.0
TOU timer + L3	697.1 [641.0, 750.9]	0.0862	0.132	0.995	0.00	0.9515	23.7
Price-aware heuristic	619.8 [561.8, 674.2]	0.0766	0.200	0.997	0.73	0.9598	0.0
Price-aware heuristic + L3	619.8 [564.4, 675.0]	0.0766	0.200	0.997	0.00	0.9623	1.7
MPC LP-OPF (perfect foresight)	612.0 [555.7, 667.2]	0.0759	0.200	0.992	0.01	0.9634	0.0
MPC LP-OPF (perfect foresight) + L3	612.0 [555.3, 667.3]	0.0759	0.200	0.992	0.00	0.9635	0.0
Flat DDPG	756.1 [695.5, 828.3]	0.0934	0.203	0.996	0.45	0.9673	0.0
Flat DDPG + L3	756.1 [694.6, 827.0]	0.0934	0.203	0.996	0.00	0.9686	0.1
PPO-Lagrangian	790.2 [725.8, 866.6]	0.0976	0.199	0.996	0.19	0.9758	0.0
PPO-Lagrangian + L3	790.2 [725.2, 865.0]	0.0976	0.199	0.996	0.00	0.9763	0.0
CPO	816.9 [747.5, 900.1]	0.1006	0.200	0.999	0.35	0.9731	0.0
CPO + L3	817.0 [748.1, 903.3]	0.1006	0.200	0.999	0.00	0.9739	0.0
Hierarchical RL	741.3 [685.3, 799.4]	0.0915	0.200	0.997	0.03	0.9741	0.0
Hierarchical RL + L3	741.3 [685.3, 800.0]	0.0915	0.200	0.997	0.00	0.9743	0.0
Nested (proposed)	645.0 [590.0, 700.0]	0.0797	0.201	0.997	0.00	0.9665	0.4

TABLE IV

DAILY ENERGY COST (€) AND VIOLATION RATE (%) IN S1–S5 (PENETRATION IN PARENTHESES; S4: S3 WITH BASE-LOAD DEVIATION). THE LEARNED BASELINES WITH LEVEL 3 (NOT SHOWN) HAVE NO VIOLATIONS IN S1, S2, S3 AND S5

Method	S1 (40%)		S2 (67%)		S3 (80%)		S4 (80%)		S5 (100%)	
	Cost	Viol.	Cost	Viol.	Cost	Viol.	Cost	Viol.	Cost	Viol.
Uncoordinated	474	0.42	780	2.35	931	3.99	931	4.53	1161	6.82
TOU timer	358	0.06	587	2.39	698	4.30	695	4.31	858	7.55
Uncoordinated + L3	474	0.00	780	0.00	931	0.00	931	0.02	1161	0.00
TOU timer + L3	358	0.00	587	0.00	697	0.00	695	0.00	858	0.00
Price-aware heuristic + L3	317	0.00	520	0.00	620	0.00	618	0.02	773	0.00
MPC LP-OPF (perfect foresight) + L3	313	0.00	513	0.00	612	0.00	610	0.00	765	0.00
Flat DDPG	409	0.09	642	0.30	756	0.45	754	0.84	931	1.11
PPO-Lagrangian	403	0.01	662	0.16	790	0.19	788	0.62	984	0.51
CPO	420	0.03	684	0.27	817	0.35	814	0.93	1016	0.81
Hierarchical RL	380	0.00	622	0.04	741	0.03	739	0.38	922	0.19
Nested (proposed)	329	0.00	540	0.00	645	0.00	642	0.00	806	0.00

TABLE V

PAIRED COMPARISON AT S3 WITH THE BASELINES WITH LEVEL 3 (Δ : NESTED MINUS BASELINE, 95% BOOTSTRAP INTERVAL; DAYS: DAYS OF 50 ON WHICH NESTED IS CHEAPER; WILCOXON TESTS, HOLM OVER 16 COMPARISONS PER METRIC)

Nested vs.	Δ Cost (€)	p_{Holm}	Days	Δ SQ	p_{Holm}
Uncoord.+L3	-286.4 [-377.8, -219.2]	<0.001	50	-0.0028	<0.001
TOU+L3	-52.0 [-71.9, -34.0]	<0.001	41	+0.0016	0.011
Price-aware+L3	+25.2 [+21.9, +28.8]	<0.001	0	+0.0001	1.000
LP-OPF*+L3	+33.0 [+29.0, +37.3]	<0.001	0	+0.0046	<0.001
Flat DDPG+L3	-111.0 [-141.9, -89.8]	<0.001	50	+0.0009	<0.001
PPO-Lag+L3	-145.2 [-179.5, -118.8]	<0.001	50	+0.0011	<0.001
CPO+L3	-171.9 [-216.2, -138.7]	<0.001	50	-0.0018	<0.001
HRL+L3	-96.2 [-108.5, -85.5]	<0.001	50	-0.0001	1.000

TABLE VI

LEVEL-3 STRESS TESTS (S6: NO CHARGING 19:00–20:00; S7: 360 VEHICLES, $S_i = 6$ kVA). VIOL. IN %; $V^{\min}$: MEAN DAILY MINIMUM VOLTAGE (P.U.); CURT. AND UNMET IN KWH PER DAY

Method	S6		S7			
	Viol.	$V^{\min}$	Viol.	Curt.	Unmet	SQ
Uncoord.	3.80	0.951	9.00	0	6	1.000
TOU	4.32	0.937	10.62	0	402	0.968
Uncoord.+L3	0.00	0.958	0.35	492	15	0.999
TOU+L3	0.00	0.951	0.17	682	561	0.955
Price-aware+L3	0.00	0.962	0.16	139	106	0.991
LP-OPF*+L3	0.00	0.963	0.00	0	98	0.992
Nested	0.00	0.966	0.01	46	38	0.997

0.33%.

attribute no cost benefit to the hourly price level. Without the curriculum the cost is lower by 1.0% (S3) and 1.2% (S5), a resolved difference. Without the behavioral model every driver accepts, which raises service quality and lowers the cost by 1.0%: price response constrains the controller. A policy trained without Level 3 violates the floor in 0.23% of the steps, and removing Level 3 from the trained controller at test time gives

G. Generalization

Table IX repeats S3 on the 69-bus feeder, on recorded ACN-Data sessions from two workplace sites, and on 2019 days with their prices (this experiment reuses the policies trained on 2023 days). In every setting the nested controller costs 14.0%–

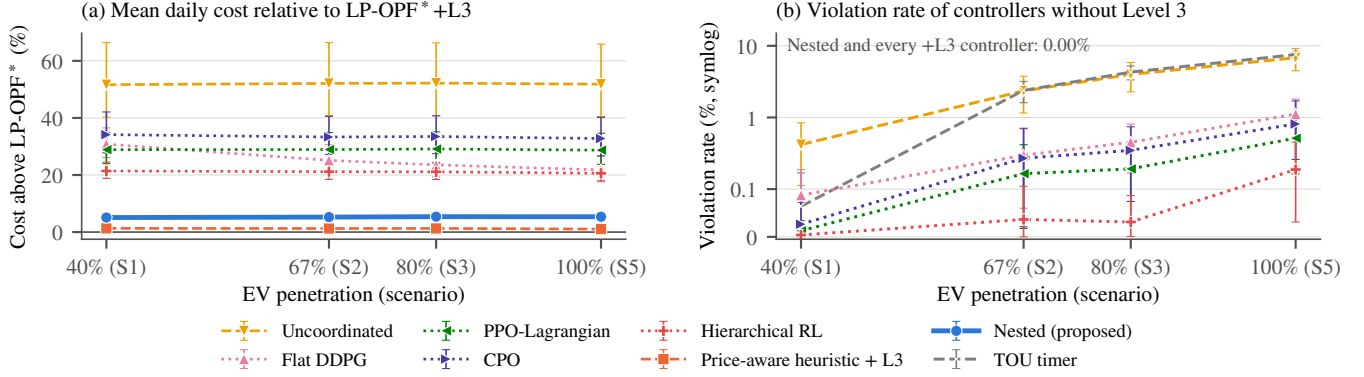


Fig. 4. (a) Mean daily energy cost relative to that of the perfect-foresight LP-OPF with Level 3 (ratio of means over the test days; 95% intervals from a bootstrap over days that keeps both methods' costs of a day together), and (b) violation rate of the controllers without Level 3 (symmetric-log axis), across S1, S2, S3 and S5. The nested controller and every "+L3" controller have no violations in these scenarios.

TABLE VII

CONTRIBUTION OF THE LEARNED LEVELS (LEVEL 3 ACTIVE). PLAN: PLANNING PRIOR WITHOUT LEARNING; PRICES ONLY: LEARNED PRICES, PLAN DISPATCH; DISPATCH ONLY: LEARNED DISPATCH, FLAT TARIFF. UNITS AS IN TABLE VI

Configuration	S3			S7			
	Cost	Retail	SQ	Cost	Viol.	Curt.	Unmet
Price-aware+L3	619.8	0.200	0.997	940.4	0.16	139	106
Plan+L3	627.9	0.200	0.997	956.7	0.02	114	37
Prices only	633.9	0.201	0.997	966.3	0.03	82	43
Dispatch only	645.3	0.200	0.998	981.4	0.03	71	30
Nested	645.0	0.201	0.997	982.9	0.01	46	38

TABLE VIII

ABLATIONS AT S3 AND S5, THREE TRAINING SEEDS EACH (COMPLETE FRAMEWORK ON THE SAME SEEDS). COST IN €, VIOL. IN %; -: NOT EVALUATED

Configuration	S3			S5		
	Cost	SQ	Viol.	Cost	SQ	Viol.
Complete framework	645.7	0.997	0.00	806.6	0.997	0.00
No planning prior	696.5	0.996	0.00	865.5	0.996	0.00
No Level 1	642.8	0.996	0.00	804.6	0.996	0.00
Single timescale	651.2	0.996	0.00	812.8	0.996	0.00
No behavior model	639.4	1.000	0.00	800.0	0.999	0.00
No Level 3 (trained)	648.5	0.996	0.23	810.5	0.995	0.71
Level 3 removed at test	645.7	0.997	0.33	-	-	-
No curriculum	639.2	0.996	0.00	797.3	0.996	0.00
No deadline guard	651.2	0.996	0.00	814.6	0.996	0.00
Proportional allocation	660.8	0.995	0.00	828.5	0.995	0.00

30.8% less than uncoordinated charging and 2.7%–6.6% more than the price-aware heuristic, with no violations. At JPL the price-aware heuristic is cheaper than the LP-OPF, which is a reference, not a bound (Section V). At the two workplace sites the floor is rarely at risk: uncoordinated charging has no violations at Caltech and violations in at most 0.005% of the steps at JPL. At Caltech every method, uncoordinated charging included, delivers a smaller share of the requested energy than in any other setting (service quality at most 0.971).

TABLE IX

GENERALIZATION AT 80% PENETRATION (THREE TRAINING SEEDS). UNC.: UNCOORDINATED; PA: PRICE-AWARE HEURISTIC; LP*: PERFECT-FORESIGHT LP-OPF

Setting	Daily cost (€)				Viol. (%)		SQ
	Unc.	PA+L3	LP*+L3	Nested	Unc.	Nested	
33-bus, residential	931.4	619.8	612.0	645.0	3.99	0.00	0.997
69-bus, residential	931.4	619.8	611.7	648.0	0.21	0.00	0.997
33-bus, ACN Caltech	146.5	121.3	119.7	126.0	0.00	0.00	0.970
33-bus, ACN JPL	263.8	194.1	197.1	207.0	0.00	0.00	0.994
33-bus, 2019 days	371.1	282.3	280.2	289.9	4.72	0.00	0.997

H. Level 3 Parameters

Varying the correction margin ($\delta = 0.0005, 0.001$ and 0.002 p.u.) and the inverter rating ($S_i = 10, 12$ and 14 kVA, where 10 kVA emulates chargers with reduced reactive capability) at S3 causes no violations and changes the cost of the nested controller by at most 0.01% . A larger margin activates Level 3 in more of the 60-s steps ($0.30\%, 0.32\%$ and 0.37%), and a smaller rating shifts the correction from reactive power to curtailment ($0.0, 0.4$ and 3.9 kWh per day at $14, 12$ and 10 kVA).

VII. DISCUSSION

A. Role of Learning

At nominal load, learning does not lower the energy bill relative to a well-designed rule: the price-aware heuristic, which executes the Level-2 plan per vehicle, is cheaper than the nested controller, and neither the learned prices nor the learned dispatch reduce cost relative to the plan (Table VII). The learned levels reduce curtailment, which mattered under stress (S7); the lower unmet energy under stress comes from least-laxity-first allocation, not from learning. Learned dispatch needs the plan as a starting point: learned from scratch, it is costlier at both penetration levels tested (S3 and S5). The nested design is therefore not a way to beat optimization on cost but a way to keep learned components where they help while planning and physics handle the rest.

B. Scope of the Voltage Correction

Level 3 is corrective control on the simulated quasi-static state, not an a priori guarantee. It restores the floor whenever the reactive headroom of the plugged-in chargers suffices, and otherwise sheds EV power until the floor is met or the iteration limit is reached (Section IV). Three consequences follow. First, the inverter apparent-power rating sets the voltage margin: Section VI-H reports the effect of S_i , and S7 the behavior when the reactive capability is halved. Second, we assume, as in [21], that a charger can supply reactive power at any active power, including when it does not charge; chargers without this capability offer only the curtailment fallback. Third, the floor is applied at the medium-voltage (12.66 kV) buses of the test feeders; drops in service transformers and secondary circuits are not modeled, and the correction acts at the 60-s resolution of the simulation, so switching transients and sub-cycle dynamics are outside its scope, as in any quasi-static study.

C. Assumptions and Limitations

Information. The plan, the learned agents and the heuristic know each plugged-in vehicle's departure time and remaining need, as in managed-charging programs where drivers enter them [24]. They do not know future arrivals. Only the LP-OPF reference sees the whole day in advance. *Network.* The feeders are balanced, radial and single-phase equivalent; unbalanced three-phase operation can change the voltage sensitivities on which the correction step relies. *Behavior.* The acceptance model (2) is stylized. The voltage layer does not depend on it, but the value of price signals does. *Fleets.* The residential fleet is synthetic with stated parameters; the ACN experiments use recorded sessions from workplace sites. *Data.* Household loads (Pecan Street, United States) and day-ahead prices (Netherlands) come from different systems; they provide realistic daily shapes, not a model of one particular utility. Training and test days share the base-load profile and differ in prices, fleets and noise. Adaptation to structural market change is not tested beyond the 2019 price regime.

VIII. CONCLUSION

We presented a nested EV charging controller with hourly pricing, 15-min plan-residual dispatch and 60-s reactive voltage correction. On two IEEE feeders with AC power flow and real price, load and session data, it kept the voltage floor without base-load deviation, up to 100% penetration on the 33-bus feeder and at 80% in the other settings, and was the cheapest learned controller. A price-aware heuristic remained cheaper at nominal load, and a perfect-foresight LP-OPF cheaper still. Under stress beyond the chargers' reactive capability, the controller curtailed and left undelivered far less energy than the heuristic; the lower curtailment came from the learned levels, the lower unmet energy from laxity-first allocation. Two lessons follow. First, a physics-based correction layer bounded by the inverter rating removed the violations of every coordinator we attached it to, up to 100% penetration without base-load deviation, at an energy-cost change of at most 0.10%; such a layer belongs in every

comparison. Second, learned dispatch is worth its complexity mainly when it starts from a sound plan and grid stress, not price, is the binding concern. Future work includes unbalanced three-phase feeders, uncertain departure times and field validation.

DATA AND CODE AVAILABILITY

Code, configuration, trained policies, evaluation logs and the script that regenerates every table, figure and number are available at <https://github.com/Abraheem13/nested-ev-grid>. All data are openly available through the EV2Gym and SustainGym packages and the MATPOWER feeder files (Section III) and are verified by SHA-256 checksums. Household load data were provided by Pecan Street Inc. (Dataport).

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Abraheem Rashid is with the Department of Computer Science, Brunel University London, London, U.K.

Faisal Iradat is with the Department of Computer Science, Institute of Business Administration, Karachi, Pakistan.

Waseem Iqbal is with the Department of Electrical and Computer Engineering, Sultan Qaboos University, Muscat, Oman.

Yawar Abbas Bangash is with the School of Computing and Artificial Intelligence, Sunway University, Selangor, Malaysia.